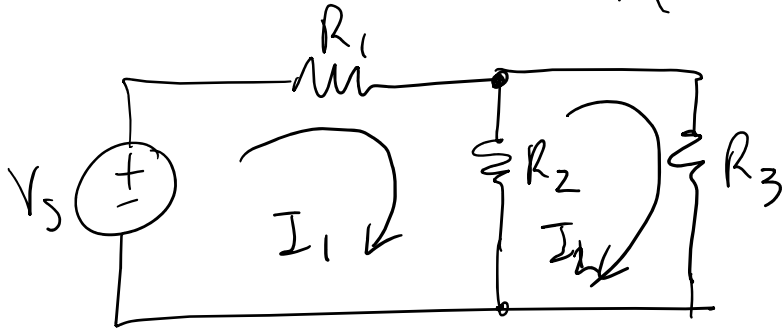
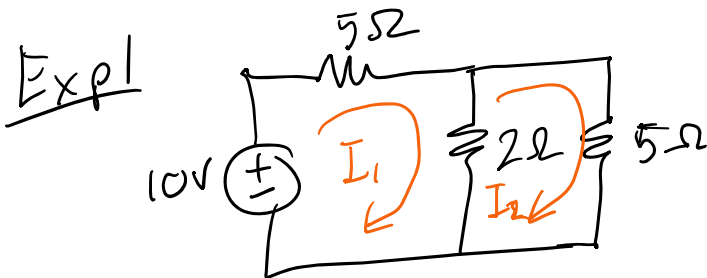
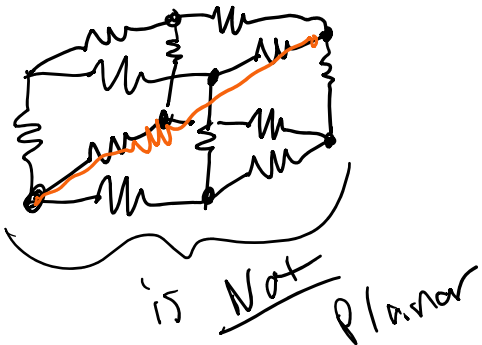


Mesh Analysis (KVL) (Loop analysis)



KVL states that the sum of voltages in a loop/mesh must equal zero.

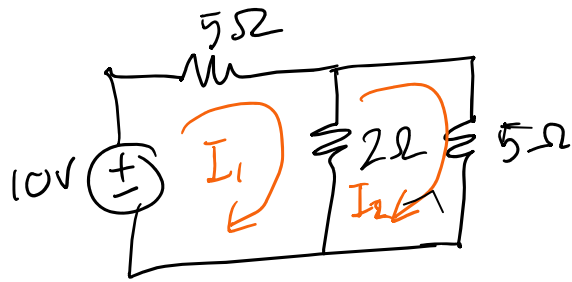
* Mesh only works on "planar" circuits



Mesh 1

$$5I_1 + 2(I_1 - I_2) - 10V = 0$$

$$7I_1 - 2I_2 = 10V$$



Mesh 2

$$2(I_2 - I_1) + 5(I_2) = 0$$

$$-2I_1 + 7I_2 = 0 \dots \textcircled{2}$$

$$\left. \begin{array}{l} 7I_1 - 2I_2 = 10 \\ -2I_1 + 7I_2 = 0 \end{array} \right\} \begin{array}{l} \text{Put} \\ \text{in} \\ \text{calculator} \end{array} \quad \boxed{\begin{array}{l} I_1 = 1.56A \\ I_2 = 0.44A \end{array}}$$

Cramer's Rule

$$\begin{array}{cc} I_1 & I_2 \\ \begin{bmatrix} 7 & -2 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix} \end{array}$$

Finding the Determinant

$$(7 \cdot 7) - (-2 \cdot -2)$$

$$(49) - (4) = 45$$

$$I_1 = \frac{\begin{bmatrix} 10 & -2 \\ 0 & 7 \end{bmatrix}}{45} \quad D = 45 \quad I_2 = \frac{\begin{bmatrix} 7 & 10 \\ -2 & 0 \end{bmatrix}}{45}$$

$$I_1 = \frac{\begin{bmatrix} 10 & -2 \\ 0 & 7 \end{bmatrix}}{45} \quad D_0 = 45$$

$$I_2 = \frac{\begin{bmatrix} 7 & 10 \\ -2 & 0 \end{bmatrix}}{45}$$

$$I_1 = \frac{(10 \cdot 7) - (-2 \cdot 0)}{45}$$

$$I_2 = \frac{(7 \cdot 0) - (10 \cdot -2)}{45}$$

$$I_1 = \frac{70}{45}$$

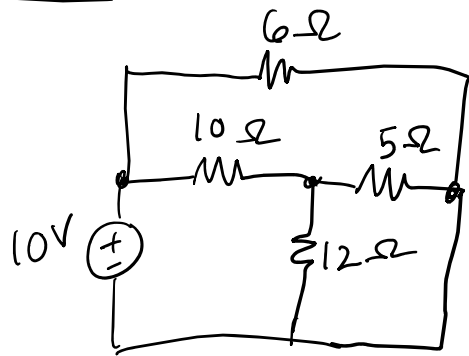
$$I_2 = \frac{20}{45}$$

$$\boxed{I_1 = 1.56A}$$

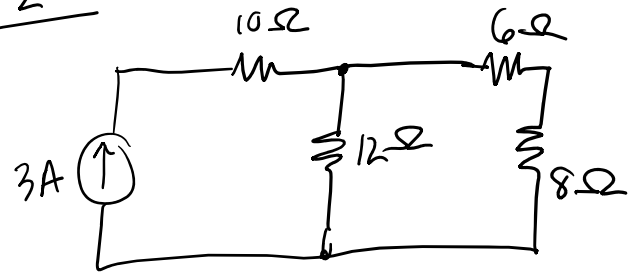
$$\boxed{I_2 = 0.44A}$$

Group Problems

Exp 1



Exp 2



Exp 3

